

Cell Biology: Structure & Function

1. Prokaryotic vs Eukaryotic

Prokaryotes (bacteria, archaea) lack a membrane-bound nucleus and organelles. Eukaryotes (animals, plants, fungi) compartmentalize functions inside organelles.

2. Key Organelles

- Nucleus: stores DNA, controls gene expression.
- Mitochondria: generate ATP via oxidative phosphorylation.
- Ribosomes: synthesize proteins from mRNA.
- Endoplasmic reticulum: rough ER folds proteins, smooth ER handles lipid synthesis.
- Golgi apparatus: modifies, sorts, and packages proteins for transport.

3. The Cell Membrane

A phospholipid bilayer with embedded proteins controls what enters and exits the cell — selectively permeable, not just a passive barrier.